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Roll No

CE-803 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Advanced Structure Design-II (Steel)
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions .
ii) All questions carry equal marks.

1. Design a silo to hold 600kN of wheat unit weight and wheat 8.5 kN/m^3 and $\phi = 25^\circ$. Sketch the details.
2. Design the container of 200kL suspended bottom overhead steel tank. Assume suitable shape for the tank.
3. Design the container of a circular hemispherical bottom water tank for capacity of 150kN located at Bhopal with following details
Staging height = 12m
Basic wind speed = 47m/sec
Seismic zone = II zone
4. Design for indore a self - supporting steel stack for Jabalpur of height 60 m above the foundation and diameter of cylindrical part is 4.25 m. Foundation is raft resting on medium soil. Assume suitable data.
5. What do you understand by Riveted connective? Also explain Rivet value.
6. a) Discuss the design of chimney base.
b) Explain the design of Guyed steel stacks.
7. A steel bunker 10m length and 8m width is required to store coal of bulk density 8000 N/m^3 . The angle of internal friction is 35° . Height of vertical portion and hopper is 4m each. Design the side walls and through portions.
8. Define the following (any two):
 - i) End bearings for steel bridges
 - ii) Type of trusses
 - iii) Lining material used for chimneys
 - iv) Analysis of towers
